

HealthConnect Clinic: Initial Analysis & Strategy

Week 4: Technical Problem Understanding and Solution Design for Patient Attendance Optimisation

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1. Executive Summary & Business Scenario

HealthConnect Clinic operates as an essential community healthcare provider delivering appointment-based medical services. Currently, the clinic faces major operational, financial, and logistical bottlenecks because of missed scheduled appointments (no-shows). When a patient fails to show up for their scheduled slot, clinical rooms sit empty, clinician time is wasted, and other community members who require urgent care are deprived of timely medical interventions. Additionally, administrative resources are heavily strained by manual follow-ups and repetitive enquiries regarding appointments and procedures.

The primary objective of this project is to leverage clinical data, predictive modelling, and generative AI to address this core problem. The central project question driving this work is:

Central Project Question:

How can HealthConnect Clinic utilise clinical and booking data alongside AI technologies to reduce missed appointments, optimise clinic operations, and improve the patient support experience?

As a Data Analyst, my responsibility is to conduct a thorough analysis of historical patient booking behaviours, identify the core underlying variables associated with no-show patterns, establish operational key performance indicators (KPIs), and present actionable, data-driven recommendations to the clinical management team to guide targeted intervention strategies.

2. Dataset Overview & Structure

The primary source of analytical evidence is the HealthConnect Appointment Dataset (HealthConnect_Appointment_Data.csv), which contains fictionalised and anonymised medical appointment records. To protect patient privacy and comply with health data protection policies, all personally identifiable information (PII) has been stripped.

The dataset's key characteristics and profile details are structured as follows:

Dataset Attribute (Metric)	Value / Details
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Dataset File Name	HealthConnect_Appointment_Data.csv
Total Volume (Records)	5,000 appointment records
Feature Space (Variables)	18 variables (columns)
Unique Patient Cohort	1,696 unique patients
Duplicate Records	0 duplicate rows detected (100% key uniqueness)

The 18 available features are structured logically across four main business domains:

- **Patient Demographics:** Includes patient_id (anonymised key), gender (Female, Male, Prefer not to say), age (ranging from 18 to 80 years), and age_group (categorised into six distinct bands).
- **Appointment Details:** Captures appointment_id (prefixed 'HC-'), appointment_type (General Consultation, Follow-up, Specialist Consultation, Diagnostic Test), appointment_date, appointment_day, and appointment_time (Morning, Afternoon, Evening).
- **Booking Information & History:** Focuses on booking_date, booking_lead_days (the lag from booking to appointment date, ranging from 0 to 60 days), previous_appointments (0 to 11 visits), and previous_no_shows (0 to 5 missed visits).
- **Logistics, Reminders & Outcome:** Records distance_to_clinic_km, waiting_time_minutes, reminder_sent (Yes, No), reminder_channel (SMS, WhatsApp, Email, or blank if unsent), and the final appointment_outcome (Attended, No-Show, Cancelled).

3. Baseline Performance & Week 4 Strategic Insights

An initial profiling of the 5,000 scheduled appointments reveals a critical operational crisis for HealthConnect Clinic: nearly half of all booked visits are missed. This high-level baseline represents a massive drain on clinical efficiency and community care capacity.

Appointment Outcome	Volume (Count)	Percentage (%)
No-Show (Missed)	2,423	48.5%
Attended (Completed)	2,314	46.3%
Cancelled	263	5.3%
Total Booked Appointments	5,000	100.0%

Early exploratory analysis indicates that several core factors drive patient no-shows. These insights provide the technical context for our analytics strategy:

- **Booking Lead Days is the Strongest Predictor:** There is a notable 10-day difference in booking lag times. Patients who successfully attended their appointments had an average booking lead time of 24.5 days, whereas those who defaulted booked their appointments 34.5 days in advance. This strongly suggests that patient engagement and intent decay over time.
- **Past Behaviour Dictates Future Action:** Default patterns are highly habit-based. No-show patients had an average historical count of 0.64 previous no-shows, compared to 0.46 for patients who attended their scheduled consultations.
- **Reminders Show Marginal Success:** Current communication channels underperform. The no-show rate for patients who received a reminder was 47.4%, compared to 51.4% for those who did not. This modest 4.0% reduction suggests that the wording, timing, or delivery channels of the current reminder framework require complete re-engineering.
- **Logistical Distance Barriers Exist:** Distance serves as a tangible obstacle to care. The average distance travelled by no-show patients was 10.5 km, compared to 9.7 km for those who attended, indicating that transportation and associated costs negatively impact patient compliance.
- **Appointment Type Variations:** Urgency affects commitment. Follow-up appointments exhibit the highest miss rate at 51.2%, whereas first-time General Consultations have lower miss rates (46.6%), signaling that patients place lower perceived value or urgency on routine post-treatment visits.

4. Data Quality Assessment Plan

To ensure analytical integrity and prevent distorted insights, a structured, non-destructive data quality assessment plan has been established. In compliance with program guidelines, all data exploration and cleaning procedures will be executed on a separate, mirrored dataset, keeping the original raw CSV completely intact.

Our baseline quality audit has flagged several minor missing data gaps and critical data-type discrepancies that must be addressed during the initial pipeline run:

Variable Name	Anomalies & Gaps Identified	Analytical Impact & Remediation Plan
reminder_channel	1,366 rows (27.3%) are null/blank.	Expected logical behavior. These records correlate perfectly with 'reminder_sent = No'. No cleanup required; will be isolated during reminder-specific studies.
distance_to_clinic_km	90 rows (1.8%) are null/blank.	Minor logistical data gap. Missing entries will be imputed using the cohort's median distance to prevent skewing spatial travel analyses.
waiting_time_minutes	60 rows (1.2%) are null/blank.	Logical operational behavior. Patients who are registered as 'No-Show' never checked in to wait. Will be excluded from clinical waiting-time calculations.

booking_date & appointment_date	Currently stored as String (Text) formats.	Technical blocker for chronological calculations. Must be converted into ISO-compliant datetime formats in Week 5 to calculate precise lead times.
Entire Dataset	Zero duplicates; ages valid (18-80).	High baseline integrity. Checked all 5,000 keys; zero row duplications. Demographics and bookings are structurally sound and range-valid.

5. Core Business Questions & Key Performance Indicators

To transform these raw observations into clinical results, we have formulated five target Business Questions (BQs), each directly mapped to a specific Key Performance Indicator (KPI). During Week 4, we define and justify these metrics; calculation and visual reporting will take place in subsequent project phases.

BQ1: What is the overall baseline attendance pattern, and how does it vary by operational factors?

- **Proposed Metric (KPI 1): Overall Appointment No-Show Rate.**

- *Mathematical Formula: $(\text{Total Scheduled No-Show Appointments} / \text{Total Scheduled Appointments}) * 100$*

- **Business Justification:** This is the clinic's headline metric. It quantifies the scale of missed opportunities and serves as the baseline to measure the success of all future administrative and AI-driven interventions. The current baseline is established at 48.5%.

BQ2: Do reminders reduce appointment defaults, and which delivery channel is most effective?

- **Proposed Metric (KPI 2): Reminder Effectiveness and Channel Conversion Rate.**

- *Mathematical Formula: $\text{No-Show Rate (Reminders Unsent)} - \text{No-Show Rate (Reminders Sent by SMS/WhatsApp/Email)}$*

- **Business Justification:** This metric measures communication ROI. It evaluates whether sending reminders delivers a statistically significant change in patient behavior and identifies which platform (SMS, WhatsApp, or Email) maximizes attendance, allowing the clinic to allocate its budget efficiently.

BQ3: Does scheduling appointments far in advance increase the probability of a no-show?

- **Proposed Metric (KPI 3): No-Show Rate by Booking Lead Time Band.**

- *Mathematical Formula: No-Show Rate calculated independently across lag groups (e.g., 0-7 days, 8-14 days, 15-30 days, 31-60 days)*

- **Business Justification:** Confirms the correlation between lead time and patient decay. This helps management decide whether to restrict booking windows, introduce short-term priority slots, or implement intensive reminder triggers as lead times exceed a critical threshold.

BQ4: Do patients with a history of missed appointments exhibit a higher risk of defaulting again?

- **Proposed Metric (KPI 4): High-Risk Patient Re-Attendance Rate.**

- *Mathematical Formula: Attendance and No-Show Rates grouped by prior default count (0, 1, 2, or 3+ historical no-shows)*

- *Business Justification: Enables proactive patient risk segmentation. By identifying chronic defaulters before their scheduled dates, the clinic can trigger high-priority phone calls, offer social support, or adapt clinical schedules to mitigate the higher risk.*

BQ5: Does physical distance to the clinic serve as a key barrier to appointment compliance?

- **Proposed Metric (KPI 5): No-Show Rate by Distance Cohort.**

- *Mathematical Formula: No-Show Rate segmented by travel distance intervals (e.g., <2 km, 2-5 km, 5-10 km, >10 km)*

- *Business Justification: Identifies geographical and socio-economic barriers. If a clear distance threshold correlates with high defaults, it provides a strong business case for introducing community shuttle services, mobile clinics, or virtual telehealth consultations.*

6. Assumptions, Limitations, Risks, & Dependencies

A professional data analysis must acknowledge its boundaries and assumptions to maintain integrity and prevent analytical bias. This section details the constraints impacting this project:

6.1 Analytical Assumptions

- **Historical Behaviour Consistency:** We assume that the historical patterns in the dataset represent future patient tendencies.
- **Outcome Recording Validity:** We assume clinical staff recorded the status of every appointment ('Attended', 'No-Show', 'Cancelled') accurately at check-in with zero human entry error.
- **Cancellation Exclusion:** We assume cancelled appointments represent patients acting responsibly. They will be excluded from the core no-show calculations so we can focus strictly on unnotified missed visits.

6.2 Data & Structural Limitations

- **Lack of Qualitative Context:** The dataset lacks qualitative fields. We do not know the exact reasons for missed appointments—such as transport costs, medical emergencies, sudden schedule changes, or forgetfulness.
- **Waiting Time Restriction:** Waiting time is only captured for patients who actually attended. We cannot measure if long past wait times discourage patients from returning.
- **Lack of Socio-economic Metrics:** Beyond travel distance, we have no indicators of income, employment, or medical aids, limiting our ability to identify broader socio-economic barriers.

6.3 Project Risks & Technical Dependencies

- **Synthetic Data Variance:** Since the data is fictional and anonymised, identified patterns may lack some real-world operational nuances.

- **Date Formatting Dependencies:** If text-based date fields are not converted to standard datetimes, lead-time calculations will fail, blocking Week 5 EDA progress.
- **Sequential Workflow Dependency:** Descriptive, exploratory, and visual phases (Phases 2 and 3) cannot begin until Phase 1 data cleaning and quality imputation are complete in Week 5.

7. Analytical Technology Stack & Week 5 Roadmap

To deliver a reproducible, high-quality analysis, we will use a standardized, industry-grade analytics stack:

- **Python (with Pandas and NumPy):** Dedicated to raw data profiling, handling null values, datetime formatting, and cohort segmentation.
- **Matplotlib and Seaborn:** To construct clear, publication-quality visualizations (bar charts, line plots, and distribution curves).
- **Jupyter Notebooks:** Used as the scratchpad to document step-by-step cleaning and exploratory code, maintaining reproducibility.
- **SQL Database Platform:** For structured, efficient querying, checking for duplicates, and generating quick operational counts.
- **Power BI:** Our target platform in the final weeks to deliver an interactive operational dashboard for clinical decision-makers.

The project will follow a structured, multi-week timeline. Our immediate, high-priority tasks as we transition into Week 5 are:

- **Phase 1:** Impute the 90 missing distance values using the cohort's median, map the blank reminder channels to 'No' reminder records, and convert booking and appointment date columns into proper datetime formats.
- **Phase 2:** Group the continuous 'booking_lead_days' variable into distinct categorical bands (e.g., 0-7, 8-14, 15-30, 31-60 days) to prepare the data for KPI calculations.
- **Phase 3:** Run comprehensive descriptive statistics to calculate overall attendance rates and segment no-show rates across demographic groups (age, gender) and logistical dimensions (distance, booking bands).
- **Phase 4:** Draft initial Matplotlib and Seaborn charts to visualize the relationship between distance, booking lead times, and no-show occurrences, building a clear narrative for our final report.